

3.9 Complex numbers

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Write $z=x+iy$; modulus and argument give polar form. Multiplication adds arguments and multiplies moduli; conjugate multiplication rationalises division. Real-coefficient polynomial equations have complex roots in conjugate pairs, a fact useful when solving quadratics and higher-degree equations.

Worked example: follow the reasoning

$(1+i)(1-i)=2$; reciprocal of $1+i$ is $(1-i)/2$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Express $(3+4i)/(1-i)$ in $a+bi$ form.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Multiply top and bottom by $(1+i)$ [1]; numerator $(3+4i)(1+i)=-1+7i$ [1]; divide 2: $-1/2+(7/2)i$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.